The main undesirable properties of cement concrete are as follows:

i. It undergoes shrinkage while setting and hardening.
ii. It requires careful attention in preparation, placing and curing.
iii. Concrete structure will be bigger and heavier than steel structures.
iv. If it is not compacted thoroughly, porous holes may formed on the surface.

4.3.9 Uses of Concrete

Concrete is used for variety of purposes such as:

i. Foundations of masonry works, especially in damp soil or under water.
ii. Terrace roofs and floors.
iii. Walls, retaining walls.
iv. Arches, dams and bridges, etc.



- The use of concrete can be traced back to ancient Egypt, where it was used as an infill material for the pyramids.
- First concrete highway was built in 1909 in Greenfield Township, which is now northwest Detroit, Michigan, USA.

4.3.10 Types of Concrete

There are many types of concrete that can be made as per requirement. In the ordinary concrete, the following distinctions are usually made.

(1) No fines concrete.
(2) High slump or self-compacting concrete.
(3) High strength concrete.
(4) High performance concrete.

4.1.10.1 No Fines Concrete



It is designed with cement, coarse aggregate and water without fine aggregates. This is generally used in mass concrete work in foundation where we want to prevent capillary rise of water.

4.1.10.2 High Slump or Self Compacting Concrete



In situations like concreting of piles, we cannot compact the concrete by external means. Similarly, in places where there is congestion of steel, we need this type of concrete. These are produced by

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